

Package ‘msPCA’

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Type Package

Title Sparse Principal Component Analysis with Multiple Principal Components

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Description Implements an algorithm for computing multiple sparse principal components of a dataset. The method is based on Cory-Wright and Pauphilet ``Sparse PCA with Multiple Principal Components" (2022) [doi:10.48550/arXiv.2209.14790](https://doi.org/10.48550/arXiv.2209.14790). The algorithm uses an iterative deflation heuristic with a truncated power method applied at each iteration to compute sparse principal components with controlled sparsity.

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`fraction_variance_explained`
Fraction of variance explained

Description

Computes the fraction of variance explained (variance explained normalized by the trace of the covariance/correlation matrix) by a set of PCs.

Usage

```
fraction_variance_explained(C, U)
```

Arguments

`C` A matrix. The correlation or covariance matrix ($p \times p$).
`U` A matrix. The matrix containing the r PCs ($p \times r$).

Value

A float.

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
mspcars <- mspca(TestMat, 2, c(4,4))
fraction_variance_explained(TestMat, mspcars$x_best)
```

`fraction_variance_explained_perPC`
Fraction of variance explained per PC

Description

Computes the fraction of variance explained (variance explained normalized by the trace of the covariance/correlation matrix) by each PC.

Usage

```
fraction_variance_explained_perPC(C, U)
```

Arguments

`C` A matrix. The correlation or covariance matrix ($p \times p$).
`U` A matrix. The matrix containing the r PCs ($p \times r$).

Value

An array.

mspca

*Multiple Sparse PCA***Description**

Returns multiple sparse principal component of a matrix using an iterative deflation heuristic.

Usage

```
mspca(
  Sigma,
  r,
  ks,
  maxIter = 200L,
  verbose = TRUE,
  feasibilityConstraintType = 0L,
  feasibilityTolerance = 1e-04,
  stallingTolerance = 1e-08,
  maxIterTPM = 20L,
  timeLimitTPM = 20L,
  restartsAfterFirstIter = 10L
)
```

Arguments

Sigma	A matrix. The correlation or covariance matrix, whose sparse PCs will be computed.
r	An integer. Number of principal components (PCs) to be computed.
ks	A list of integers. Target sparsity of each PC.
maxIter	(optional) An integer. Maximum number of iterations of the algorithm. Default 200.
verbose	(optional) A Boolean. Controls console output. Default TRUE.
feasibilityConstraintType	(optional) An integer. Type of feasibility constraints to be enforced. 0: orthogonality constraints; 1: uncorrelatedness constraints. Default 0.
feasibilityTolerance	(optional) A float. Tolerance for the violation of the orthogonality constraints. Default 1e-4
stallingTolerance	(optional) A float. Controls the objective improvement below which the algorithm is considered to have stalled. Default 1e-8
maxIterTPM	(optional) An integer. Maximum number of random restarts of the truncated power method (inner iteration) for the first outer iteration. Default 20.
timeLimitTPM	(optional) An integer. Maximum time in seconds for the truncated power method (inner iteration). Default 20.
restartsAfterFirstIter	(optional) An integer. Maximum number of random restarts of the truncated power method (inner iteration) for outer iterations ≥ 2 . Default 10.

Value

An object with 4 fields: 'x_best' (p x r array containing the sparse PCs), 'objective_value', 'feasibility_violation', 'runtime'.

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
mspca(TestMat, 2, c(4,4))
```

orthogonality_violation

Orthogonality constraint violation

Description

Computes the orthogonality constraint violation defined as the distance (infinity norm) between $U^T U$ and the identity matrix.

Usage

```
orthogonality_violation(U)
```

Arguments

U A matrix. Each column correspond to an p-dimensional PC.

Value

A float.

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
mspcares <- mspca(TestMat, 2, c(4,4))
orthogonality_violation(mspcares$x_best)
```

pairwise_correlation *Pairwise correlation*

Description

Computes the pairwise correlations between PCs defined as $u_t^T C u_s$.

Usage

```
pairwise_correlation(C, U)
```

Arguments

- C A matrix. The correlation or covariance matrix ($p \times p$).
- U A matrix. Each column correspond to an p -dimensional PC.

Value

A float matrix ($r \times r$).

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
mspcares <- mspca(TestMat, 2, c(4,4))
pairwise_correlation(TestMat, mspcares$x_best)
```

print_mspca	<i>Print mspca output</i>
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Description

Displays the output of the msPCA algorithm.

Usage

```
print_mspca(sol_object, C)
```

Arguments

- sol_object A list. The output of the mspca or twp function.
- C A matrix. The correlation or covariance matrix ($p \times p$).

Value

None. Prints output to console.

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
mspcares <- mspca(TestMat, 2, c(4,4))
print_mspca(mspcares, TestMat)
```

tpm	<i>Truncated Power Method</i>
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Description

Returns the leading sparse principal component of a matrix using the truncated power method.

Usage

```
tpm(Sigma, k, maxIter = 200L, verbose = TRUE, timeLimit = 10L)
```

Arguments

Sigma	A matrix. The correlation or covariance matrix, whose sparse PCs will be computed.
k	An integer. Target sparsity of the PC.
maxIter	(optional) An integer. Maximum number of iterations of the algorithm. Default 200.
verbose	(optional) A Boolean. Controls console output. Default TRUE.
timeLimit	(optional) An integer. Maximum time in seconds. Default 10.

Value

An object with 3 fields: 'x_best' (p x 1 array containing the sparse PC), 'objective_value', 'runtime'.

References

Yuan, X. T., & Zhang, T. (2013). Truncated power method for sparse eigenvalue problems. *The Journal of Machine Learning Research*, 14(1), 899-925.

Examples

```
library(datasets)
TestMat <- cor(datasets::mtcars)
tpm(TestMat, 4)
```

variance_explained_perPC	<i>Variance explained per PC</i>
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Description

Computes the variance explained by each PC.

Usage

```
variance_explained_perPC(C, U)
```

Arguments

- | | |
|---|--|
| C | A matrix. The correlation or covariance matrix ($p \times p$). |
| U | A matrix. The matrix containing the r PCs ($p \times r$). |

Value

An array.

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